

Anuj Pokhrel

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Research Interests

I'm a PhD student, advised by Dr. Xuesu Xiao. I am interested in developing robust perception and planning algorithms and applying Deep Learning/Machine Learning for effective maneuverability in challenging off-road terrain.

Education

George Mason University

Ph.D. student in Computer Science

Fairfax, VA

Jan 2021 – Dec 2026 (Expected)

- **Research Focus:** High Speed Off Road Navigation in Wheeled Robots

George Mason University

Master of Science in Computer Science

Fairfax, VA

May 2025

- **Relevant Courses:** Machine Learning, Autonomous Robotics, Analysis of Algorithms, Introduction to Artificial Intelligence, Theory of Computation, Theory/Application Data Mining.

Tribhuvan University

Bachelors in Computer Engineering

Kathmandu, Nepal

Sep 2017

Publications

PEER-REVIEWED PAPERS

Anuj Pokhrel, Aniket Datar, Mohammad Nazeri, and Xuesu Xiao. **CAHSOR: Competence-Aware High-Speed Off-Road Ground Navigation in SE(3)**. *Robotics and Automation Letters (RA-L)*, 2024.

Anuj Pokhrel, Aniket Datar, and Xuesu Xiao. **Dom, cars don't fly!—Or do they? In-Air Vehicle Maneuver for High-Speed Off-Road Navigation**. *International Conference on Intelligent Robots and Systems (IROS)*, 2025.

Aniket Datar, **Anuj Pokhrel**, Mohammad Nazeri, Madhan B. Rao, Chenhui Pan, Yufan Zhang, André Harrison, Maggie Wigness, Philip R. Osteen, Jinwei Ye, and Xuesu Xiao. **M2P2: A Multi-Modal Passive Perception Dataset for Off-Road Mobility in Extreme Low-Light Conditions**. *International Conference on Intelligent Robots and Systems (IROS)*, 2025.

Mohammad Nazeri, **Anuj Pokhrel**, Alexandyr Card, Aniket Datar, Garrett Warnell, and Xuesu Xiao. **VertiFormer: A Data-Efficient Multi-Task Transformer for Off-Road Robot Mobility**. (Under Review).

Amirreza Payandeh, **Anuj Pokhrel**, Daeun Song, Marcos Zampieri, and Xuesu Xiao. **Narrate2Nav: Real-Time Visual Navigation with Implicit Language Reasoning in Human-Centric Environments**. (Under Review).

Mohammad Nazeri, Aniket Datar, **Anuj Pokhrel**, Chenhui Pan, Garrett Warnell, and Xuesu Xiao. **Verticoder: Self-Supervised Kinodynamic Representation Learning on Vertically Challenging Terrain**. *International Conference on Robotics and Automation*, 2025.

Chenhui Pan, Aniket Datar, **Anuj Pokhrel**, Matthew Choulas, Mohammad Nazeri, and Xuesu Xiao. **Traverse the Non-Traversable: Estimating Traversability for Wheeled Mobility on Vertically Challenging Terrain**. 2025 (Under Review).

Aniket Datar, Chenhui Pan, Mohammad Nazeri, **Anuj Pokhrel**, and Xuesu Xiao. **Terrain-Attentive Learning for Efficient 6-DoF Kinodynamic Modeling on Vertically Challenging Terrain**. *International Conference on Intelligent Robots and Systems (IROS)*, 2024.

Tong Xu, Chenhui Pan, Madhan B. Rao, Aniket Datar, **Anuj Pokhrel**, Yuanjie Lu, and Xuesu Xiao. **Verti-bench: A general and scalable off-road mobility benchmark for vertically challenging terrain**. *Robotics: Science and Systems (RSS)*, 2025.

Research & Work Experience

George Mason University RobotiXX Lab

Graduate Research Assistant

Fairfax, VA, USA

Jan 2023 - Present

- Advisor: Dr. Xuesu Xiao
- Projects:
 - Led the development of CAHSOR, a high-speed off-road navigation system using a learned 6-DOF kinematic model to reason about maneuver consequences using a self-supervised multimodal vehicle-terrain interaction model, to capture complex off-road dynamics.
 - Designed, engineered, and deployed an autonomous 1/10th-scale Losi Buggy platform, integrating multiple sensors to deploy airborne model and planner to enable in-air pose control using only throttle and steering.
 - Designed and built a multi-modal sensor suite (thermal, event, stereo RGB, GPS, IMU, LiDAR) for low light navigation. Leveraged the 10-hour dataset collected for obstacle avoidance, achieving successful autonomous traversal of a 3.6km off-road trail.

George Mason University

Fairfax, VA, USA

Graduate Teaching Assistant

- Teaching Assistant:
 - (Spring 2021) Introduction to Low-Level Programming (CS 262) & Essentials of Computer Science (CS 110)
 - (Fall 2021) Introduction to Cryptography (CS587/CS487)
 - (Spring 2022) Blockchains and Cryptofinance (CS 499)
 - (Fall 2023) Secure Programming and Systems (CS 468)
 - (Fall 2024) Introduction to Data Mining (CS 584)

Los Alamos National Laboratory

Los Alamos, NM, USA

Intern: Applied Machine Learning School

Jun 2022 - Aug 2022

- Developed a semi-supervised BERT-based Named Entity Recognition (NER) model for explainable topic analysis of materials science research. The model analyzed titles of published papers, extracting key entities and enabling researchers to understand research trends in a more interpretable way. Additionally, a web application was built to facilitate user interaction and efficient extraction of NERs from the dataset, using the developed model in the backend.

Tekvortex Pvt. Ltd.

Lalitpur, Nepal

Full Stack Software Developer

Jan 2018 - Dec 2020

- Projects:
 - Worked on the design and development of CloudChomp, a web application that streamlined the AWS migration for organizations, utilizing Ruby on Rails for the back-end, AngularJS for the front-end, and a Postgres database.
 - Designed, developed, and implemented custom Salesforce applications for various organizations, leveraging my expertise in Apex, Visualforce, and Lightning Web Components.

Professional Service

PEER REVIEWER

IEEE Transactions on Robotics (T-RO)

IEEE Transactions on Field Robotics (T-FR)

Conference on Robot Learning (CoRL)

IEEE Robotics and Automation Letters (RA-L)

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

Languages

English Professional proficiency

Nepali Native proficiency

Hindi Medium proficiency

References

Dr. Xuesu Xiao (xiao@gmu.edu)

Assistant Professor

Dept. of Computer Science (CS)

George Mason University

4400 University Dr, Fairfax, VA 22030

Dr. Manish Bhattarai (ceodpspectrum@lanl.gov)

Staff Scientist

Los Alamos National Laboratory

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